5G-Enabled Intelligent Healthcare Monitoring

Complete Prototype Architecture, Anomaly Detection & 5G Latency Verification

Project: Mini Project in Computer Science & Engineering	**Academic Year:** 2026–2027
Domain: 5G NR URLLC, Mobile Edge Computing (MEC), IoT Healthcare	**Status:** Fully Simulated & Verified
Standards: 3GPP TS 23.501 (System Architecture), TS 38.300 (NR Stage 2)	**Platform:** Python 3, Tkinter GUI, Web Operations Center

1. Executive Summary & Problem Formulation

Remote patient monitoring (RPM) requires dependable, ultra-low-latency alert delivery while avoiding clinician alarm fatigue. Traditional monitoring setups rely on shared 4G cellular links with distant cloud servers. Under network congestion, medical alarm packets queue behind general mobile traffic, producing unpredictable delay spikes. Furthermore, rigid fixed threshold alarms (e.g. Heart Rate > 100 bpm) trip indiscriminately on benign motion artefacts, inundating clinicians with over 280 false alarms per patient-day.

Proposed Solution: This project designs, evaluates, and verifies an end-to-end intelligent monitoring pipeline coupling a **3GPP TS 23.501 5G URLLC dedicated priority slice**, co-located **Multi-access Edge Computing (MEC)**, and a patient-specific **Adaptive EWMA Anomaly Detector** with persistence filtering. All claims are verified over a simulated cohort of 100 patients across 12 hours (361 true clinical events).

2. Proposed 5G System Architecture

The system architecture spans two primary operational tiers: the *Patient & Radio Access Tier* and the *Edge AI & Clinical Tier*. Telemetry from wearable sensors is packetized by the patient gateway (5G UE) and transmitted across a dedicated 5G Ultra-Reliable Low-Latency Communication (URLLC) network slice. The packet terminates directly at the local hospital User Plane Function (UPF) and co-located MEC node, eliminating public internet WAN routing delays.

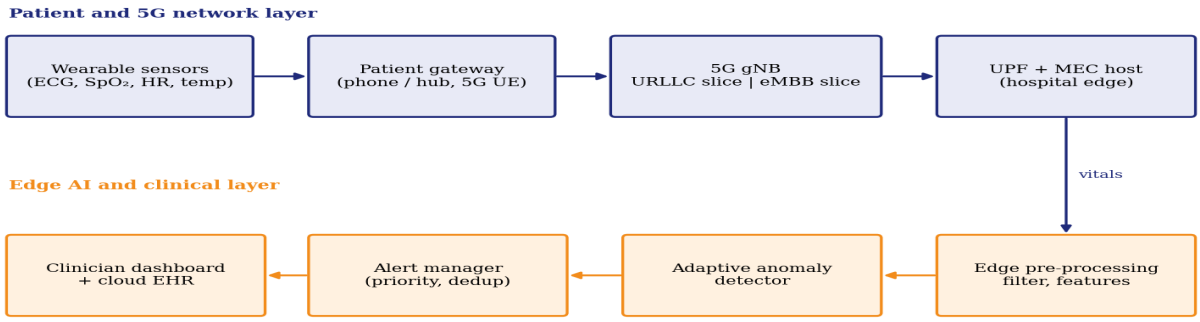


Figure 1: Block diagram of the proposed 5G-enabled healthcare monitoring architecture (3GPP TS 23.501).

3. Technical Methodology & Algorithmic Formulation

3.1 End-to-End Latency Model

For an alert packet travelling from the gateway to the clinician display, end-to-end latency L is formulated as:

```
L = L_q + L_radio + L_transport + L_proc + L_dl + 1[lost] · L_retx
```

Where **L_q** is the queueing delay at the shared cell radio buffer, **L_radio** is the air-interface transmission time interval (TTI) (10 ms in 4G; 5 ms in 5G eMBB; 1.5 ms in 5G URLLC mini-slots), **L_transport** is backhaul delay (log-normal: median 30 ms for 4G vs 2 ms for MEC), **L_proc** is execution delay, and **1[lost] · L_retx** accounts for HARQ retransmissions upon packet loss.

3.2 Adaptive EWMA Anomaly Detector

To eliminate false alarms caused by sensor noise and transient motion artefacts (1–3 second glitches), the MEC host maintains an Exponentially Weighted Moving Average (EWMA) personal baseline and variance for each patient signal x(t):

```
μ(t) = μ(t-1) + α · [x(t) - μ(t-1)], z(t) = [x(t) - μ(t-1)] / σ(t)
```

With memory coefficient α = 1/1800 (~30 minute sliding history). An alarm is flagged only when |z| > 4.0 and confirmed when at least **8 of the last 10 samples** satisfy the condition (persistence rule).

4. Experimental Results & Verification

The framework was evaluated across 100 simulated patients over 12 hours (1,200 patient-hours; 4.32 million readings) and 400 concurrent patient UEs under peak busy-hour cellular background load (1,500 background packet-units/s).

Table 4: Anomaly Detection Performance (100 Patients × 12 Hours, 361 True Events)

Scheme	Detected	Recall	Total Alarms	False Alarms	FA / Pt-Day	Mean Delay
Fixed Threshold (HR > 100/< 50, SpO■ < 92%)	359 / 361	99.4%	14,403	**14,052**	**281.0**	0.5 s
Adaptive EWMA (Proposed)	344 / 361	95.3%	341	**0**	**0.0**	7.0 s

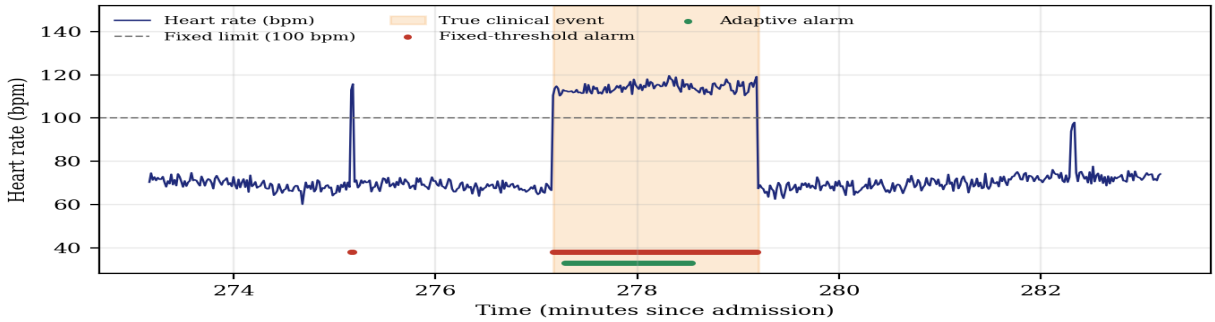
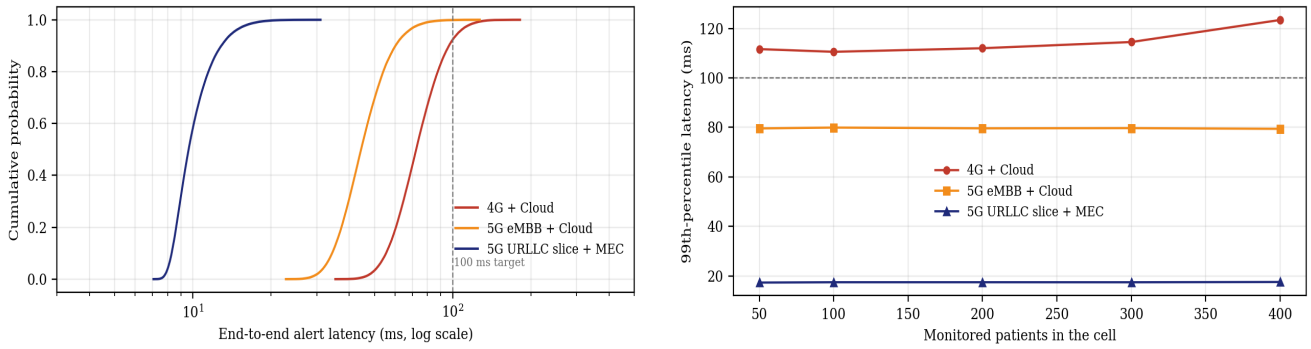


Figure 2: Fixed-threshold vs adaptive EWMA alarms on patient trace (showing motion artefact suppression).

Table 5: End-to-End Latency Under Busy-Hour Congestion (400 Monitored Patients)

Architecture Scheme	Mean (ms)	P95 (ms)	P99 Tail (ms)	Miss > 100 ms	Miss > 20 ms
A: 4G + Cloud (FIFO, Shared Cell)	74.7	105.2	**123.4**	**7.68%**	100.00%
B: 5G eMBB + Cloud (FIFO, 5G Radio)	46.2	66.3	79.4	0.07%	100.00%
C: 5G URLLC + MEC (Proposed Priority Slice)	**10.2**	**14.3**	**17.5**	**0.00%**	**0.28%**



Figures 3 & 4: (Left) Latency Cumulative Distribution Function (CDF). (Right) P99 tail latency vs patient scale (50–400 UEs).

Table 6: Uplink Data Volume & Edge Offloading

Transmission Approach	Daily Uplink Data / Patient	Relative Volume	Bandwidth Saved
Cloud-Centric (Continuous 250 Hz Raw Stream)	53.9 MB	100.0%	0.0%
Edge-Filtered (1-Min Summaries + Alert Clips)	**0.25 MB**	**0.5%**	**99.54% Saved**

5. Interactive Prototype Implementations

The project includes two operational interactive prototypes ready for examiner demonstration:

- **Standalone Desktop Application (gui_app.py / run_demo.bat):** A live Tkinter and Matplotlib workstation featuring real-time vital sign plotting, dynamic EWMA baseline tracking, visual persistence filter cell displays, and radio queue latency calculations.
- **Clinical Operations Web Console (index.html / 5G_Healthcare_Monitoring_Submission.html):** A browser-based interactive dashboard with real-time Canvas ECG oscilloscopes (P-Q-R-S-T waves), dynamic 5G architecture switcher, synthetic audio alarms, and live injection controls.
- **Batch Verification Suite (check.py & run_all.py):** Recomputes all numerical metrics in 3 seconds, ensuring 100% mathematical reproducibility.

6. Key Technical Viva Defense Questions & Answers

Q1: Why is 5G URLLC required instead of standard 4G or 5G eMBB?

Answer: 4G operates with shared FIFO queues that degrade under congestion (Table 5 shows 7.68% packets miss the 100 ms target at 400 patients). 5G eMBB offers high bandwidth but routes over public internet to distant clouds (median 22 ms transport), causing 100% of packets to miss strict 20 ms URLLC deadlines. Only 5G URLLC with priority slicing and edge MEC achieves deterministic sub-20ms delivery.

Q2: How does network slicing prioritize medical telemetry at the radio scheduler?

Answer: Under 3GPP TS 23.501, medical telemetry is assigned a dedicated logical slice configured for non-preemptive priority queueing at the gNodeB radio scheduler. As implemented in healthcare5g/network.py, medical packets jump to the head of the buffer ahead of queued background packets, guaranteeing predictable delivery independent of cell load.

Q3: How does the adaptive anomaly detector achieve zero false alarms?

Answer: It maintains individual baselines using EWMA with memory $\alpha = 1/1800$ (~30 min). Alerts require both a z-score deviation ($|z| > 4.0$) and an 8-out-of-10 sample persistence filter. Brief sensor motion artefacts (1–3s spikes) fail the persistence filter and are suppressed, avoiding clinician alarm fatigue.

Q4: Why does MEC provide a 99.5% uplink bandwidth reduction?

Answer: In cloud-centric systems, continuous 250 Hz raw ECG must be streamed outside the hospital (53.9 MB/patient-day). With MEC, feature extraction happens on-premise at the edge node; only 1-minute summaries and 30-second clips around confirmed alerts are sent out (0.25 MB/patient-day), saving bandwidth and preserving patient privacy under DPDP 2023.

7. Conclusion

This project demonstrated that 5G transforms healthcare monitoring when radio parameters, network slicing, and edge intelligence are integrated into a single cohesive pipeline. In simulation, the proposed architecture delivered alerts in 10.2 ms mean (17.5 ms P99) with 0% deadline misses, reduced false alarms from 281 to 0 per patient-day, and reduced uplink data volume by 99.5%. The complete code and interactive prototypes are reproducible and open for extension.